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SEMICONDUCTOR DIE HAVING A VARIABLE THICKNESS DEVICE LAYER

Abstract

A semiconductor die includes: a silicon-on-insulator (SOI) substrate having a silicon device layer, a bulk silicon substrate, and a buried oxide layer separating the silicon device layer from the bulk silicon substrate; a lateral power MOSFET (metal-oxide-semiconductor field-effect transistor) in a first device region of the silicon device layer; and an additional semiconductor device in a second device region of the silicon device layer and having a lower breakdown voltage than the lateral power MOSFET. The silicon device layer has a first thickness in a first part of the first device region and a second thickness in a second part of the first device region, the second thickness being greater than the first thickness. The silicon device layer has the first thickness throughout the second device region. Additional semiconductor die embodiments are also described.

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Background/Summary

BACKGROUND

[0001] Silicon-on-insulator (SOI) technology involves the fabrication of silicon semiconductor devices in a layered silicon-insulator-silicon substrate. Achieving very high breakdown voltage (e.g., >600V) using a lateral SOI complementary metal-oxide-semiconductor (CMOS) process requires a well-controlled, thin top Si (silicon) device layer thickness. However, such a very thin Si device layer thickness in SOI technology has disadvantages such as high series resistance for high voltage devices (e.g., >600V) with very long drift lengths. Additionally, the performance of low voltage devices (e.g., <600V) formed in the same Si device layer as high voltage devices may not be acceptable for low-voltage logic circuits.

[0002] Thus, there is a need for an improved SOI process that can accommodate both high-voltage devices and low-voltage devices in the same Si device layer.

SUMMARY

[0003] According to an embodiment of a method of producing power semiconductor devices from a silicon-on-insulator (SOI) wafer that includes a silicon device layer, a bulk silicon wafer, and a buried oxide layer separating the silicon device layer from the bulk silicon wafer, the method comprises: forming a hard mask on the silicon device layer, wherein the hard mask covers one or more first regions of the silicon device layer and exposes one or more second regions of the silicon device layer; and before forming any field oxide structures and before implanting any device regions, selectively growing epitaxial silicon on the one or more second regions of the silicon device layer exposed by the hard mask such that the thickness of the one or more second regions is increased relative to the one or more first regions.

[0004] According to an embodiment of a semiconductor die, the semiconductor die comprises: a silicon-on-insulator (SOI) substrate that includes a silicon device layer, a bulk silicon substrate, and a buried oxide layer separating the silicon device layer from the bulk silicon substrate; a lateral power MOSFET (metal-oxide-semiconductor field-effect transistor) formed in a first device region of the silicon device layer; and an additional semiconductor device formed in a second device region of the silicon device layer and having a lower breakdown voltage than the lateral power MOSFET, wherein the silicon device layer has a first thickness in a first part of the first device region and a second thickness in a second part of the first device region, the second thickness being greater than the first thickness, wherein the silicon device layer has the first thickness throughout the second device region.

[0005] According to an embodiment of a semiconductor die, the semiconductor die comprises: a silicon-on-insulator (SOI) substrate that includes a silicon device layer, a bulk silicon substrate, and a buried oxide layer separating the silicon device layer from the bulk silicon substrate; and a lateral JFET (junction field-effect transistor) formed in the silicon device layer, wherein a first part of the silicon device layer has a first thickness and a second part of the silicon device layer has a second thickness greater than the first thickness, wherein, in the silicon device layer, the lateral JFET comprises a source region, a drain region, a drift zone adjoining the drain region, and a body region

separating the source region from the drift zone, wherein the drift zone is formed in the second part of the silicon device layer.

[0006] Those skilled in the art will recognize additional features and advantages upon reading the following detailed description, and upon viewing the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0007] The elements of the drawings are not necessarily to scale relative to each other. Like reference numerals designate corresponding similar parts. The features of the various illustrated embodiments can be combined unless they exclude each other. Embodiments are depicted in the drawings and are detailed in the description which follows.

[0008] FIGS. 1A through 1D illustrate cross-sectional views of part of an SOI wafer during different stages of an SOI process, according to an embodiment.

[0009] FIGS. 2A through 2D illustrate cross-sectional views of part of an SOI wafer during different stages of an SOI process, according to an embodiment.

[0010] FIG. 3 illustrates a plan view of an embodiment of a lateral power MOSFET or JFET produced from the SOI wafer processed in accordance with FIGS. 1A through 1D or FIGS. 2A through 2D.

[0011] FIG. 4 illustrates a plan view of another embodiment of a lateral power MOSFET or JFET produced from the SOI wafer processed in accordance with FIGS. 1A through 1D or FIGS. 2A through 2D.

[0012] FIG. 5 illustrates cross-sectional views of an embodiment of monolithically integrating a low-voltage lateral SOI MOS transistor and a high-voltage diode in the same semiconductor die using the SOI process shown in FIGS. 1A through 1D or FIGS. 2A through 2D.

[0013] FIG. 6 illustrates a plan view of an embodiment of a power transistor device included in a semiconductor die produced using the SOI process shown in FIGS. 1A through 1D or FIGS. 2A through 2D.

[0014] FIG. 7 illustrates a plan schematic view of an embodiment of a semiconductor die produced using the SOI process shown in FIGS. 1A through 1D or FIGS. 2A through 2D and having one or more low-voltage areas and one or more high-voltage areas.

DETAILED DESCRIPTION

[0015] The embodiments described herein provide an SOI (silicon-on-insulator) process that can accommodate both high breakdown voltage devices (e.g., >600V) and low breakdown voltage devices (e.g., <600V) in the same Si device layer. The SOI process involves fabricating devices in a Si device layer with at least two different thicknesses. For example, an SOI base wafer may be provided and a hard mask such as an LPCVD (low-pressure chemical vapor deposition) oxide-based hard mask may be used to define an area where a thicker Si region is desired. The unmasked area may correspond to, e.g., the drift zone of a lateral power semiconductor device. The thickness of the Si device layer is increased in the unmasked area by selective Si epitaxial growth which is performed before forming any field oxide structures and before implanting any device regions in the Si device layer. Accordingly, existing front-end processes such as field oxide (FOX) isolation, contact etching, etc., may be used without modification.

[0016] Described next with reference to the figures are embodiments of the SOI process and devices produced using the SOI process.

[0017] FIGS. 1A through 1D illustrate cross-sectional views of part of an SOI wafer **100** during different stages of the SOI process. The SOI wafer **100** includes a silicon device layer **102**, a bulk silicon wafer **104**, and a buried oxide layer **106** separating the silicon device layer **102** from the bulk silicon wafer **104**. In an embodiment, the silicon device layer **102** initially has a single

thickness T_{Si_init} in a range of 100 to 200 nm (nanometers).

[0018] The SOI wafer **100** may be produced by SIMOX (separation by implantation of oxygen), wafer bonding, a seed method where the silicon device layer **102** is grown directly on the oxide layer **106**, etc. The part of the SOI wafer **100** shown in FIGS. **1A** through **1D** corresponds to a high breakdown voltage (e.g., >600V) device region of a semiconductor die to be produced from the SOI wafer **100**. For example, the high breakdown voltage device region may include a power semiconductor device such as a lateral power MOSFET (metal-oxide-semiconductor field-effect transistor), a lateral JFET (junction field-effect transistor), a PMOS device having a breakdown voltage greater than 600V, a power diode, etc. The SOI wafer **100** may yield tens, hundreds, or more semiconductor dies of the same design. The semiconductor dies may also include a low breakdown voltage (e.g., <600V) device region where the thickness of the silicon device layer **102** is not increased by the SOI process.

[0019] FIG. **1A** shows the SOI wafer **100** after a hard mask **108** is formed on the silicon device layer **102**. In an embodiment, the hard mask **108** comprises tetraethoxysilane (TEOS) formed by low-pressure chemical vapor deposition (LPCVD).

[0020] FIG. **1B** shows the SOI wafer **100** after patterning of the hard mask **108**. The hard mask **108**, after patterning, covers one or more first regions **110** of the silicon device layer **102** and exposes one or more second regions **112** of the silicon device layer **102**. In FIGS. **1A** through **1D**, the one or more first regions **110** of the silicon device layer **102** covered by the hard mask **108** correspond to contact regions of power semiconductor devices to be formed in the silicon device layer **102**. Some first regions **110** may correspond to field oxide regions of the power semiconductor devices and/or additional semiconductor devices having a lower breakdown voltage than the power semiconductor devices to be formed in the silicon device layer **102**. Various embodiments of the one or more first regions **110** of the silicon device layer **102** covered by the hard mask **108** are described in more detail later herein.

[0021] In an embodiment, the hard mask **108** is patterned by forming a patterned photoresist **114** on the one or more first regions **110** of the silicon device layer **102** and removing the exposed part of the hard mask **108** from the one or more second regions **112** of the silicon device layer **102**. The exposed part of the hard mask **108** may be removed by etching, for example.

[0022] In FIGS. **1A** through **1D**, the one or more second regions **112** of the silicon device layer **102** exposed by the hard mask **108** correspond to drift zones of lateral power semiconductor devices to be formed in the silicon device layer **102**. Various embodiments of the one or more second regions **112** of the silicon device layer **102** exposed by the hard mask **108** are described in more detail later herein.

[0023] FIG. **1C** shows the SOI wafer **100** after epitaxial silicon **116** is selectively grown on the one or more second regions **112** of the silicon device layer **102** exposed by the hard mask **108** such that the thickness T_{Si_epi} of the one or more second regions **112** is increased relative to the initial thickness T_{Si_init} of the silicon device layer **102**. Accordingly, the one or more first regions **110** of the silicon device layer **102** protected by the hard mask **108** during the selective epitaxial growth process are thinner than the one or more second regions **112** of the silicon device layer **102** which are subjected to the selective epitaxial growth process. The selective growth of the epitaxial silicon **116** is performed before forming any field oxide structures and before implanting any device regions in the silicon device layer **102**. In an embodiment, the selective growth of the epitaxial silicon **116** is performed by reduced pressure chemical vapour deposition (RPCVD) of silicon epitaxy.

[0024] The increased thickness T_{Si_epi} of the one or more second regions **112** of the silicon device layer **102** enables a lower R_{dson} (on-state resistance) for high breakdown voltage (e.g., >600V) lateral power semiconductor devices to be produced from the SOI wafer **100**. The R_{dson} of such devices is dominated by the thickness of the silicon device layer **102** because of the (lateral) drift zone length (e.g., 30 to 100 nm). Increasing the thickness of the silicon device layer **102** in one

or more unmasked regions **112** of the SOI wafer **100** enables R_{dson} reduction for lateral power semiconductor devices.

[0025] Other techniques for reducing R_{dson} of a lateral power semiconductor device are not as advantageous for thin SOI wafers. For example, RESURF doping optimization is limited because of the thin (e.g., 100 to 200 nm) silicon device layer **102**. Also, there is no need to redesign pre-existing front end processes such as structure topology, implants, etc. Thermal budgets also are not affected, since the epitaxy process shown in FIG. **1C** is performed early in the process before forming any field oxide structures and before implanting any device regions in the silicon device layer **102**. This allows for the device design process to be simplified by simple layout engineering. Since the hard mask **108** may be patterned as desired, multiple device options with and without increased Si epitaxy thickness may be accommodated by patterning the hard mask **108** in a corresponding manner such that only those region(s) **112** of the silicon device layer **102** that benefit from increased epitaxy thickness are exposed during the selective epitaxial growth process.

[0026] One or more regions **110** of the SOI wafer **100** that do not stand to benefit from an increased Si epitaxy thickness are protected from the Si epitaxial growth process by the patterned hard mask **108**. As explained above, the one or more masked regions **110** of the silicon device layer **102** may correspond to contact regions and/or field oxide regions of the lateral power semiconductor devices to be produced from the SOI wafer **100**, and/or to additional semiconductor devices having a lower breakdown voltage than the lateral power semiconductor devices and to be formed in the silicon device layer **102**. Accordingly, high breakdown voltage (e.g., >600V) lateral power semiconductor devices and low breakdown voltage (e.g., <600V) devices may be monolithically integrated using the same SOI wafer **100** without having to modify existing front-end processes such as field oxide isolation, contact etching, etc.

[0027] FIG. **1D** shows the SOI wafer **100** after forming field oxide regions **118** which provide intra and/or inter device isolation. In an embodiment, the field oxide regions **118** may be STI (shallow trench isolation) or LOCOS (local oxidation of silicon) structures. For example, the field oxide regions **118** may be formed by growing a field oxide over targeted region(s) of the SOI wafer **100**. If a field oxide region **118** is formed in any of the one or more second regions **112** of the silicon device layer **102** having the increased epitaxy thickness $T_{\text{Si_epi}}$, a current leakage path may arise under the field oxide region **118** if the field oxide region **118** does not extend down to the buried oxide layer **106**, as indicated by the dashed arrow in FIG. **1D**.

[0028] FIGS. **2A** through **2D** illustrate cross-sectional views of part of the SOI wafer **100** during different stages of the SOI process, according to another embodiment. The embodiment shown in FIGS. **2A** through **2D** is similar to the embodiment shown in FIGS. **1A** through **1D**. Accordingly, only the differences between the two embodiments are described here. As explained above, a current leakage path may arise under a field oxide region **118** formed in a second region **112** of the silicon device layer **102** with the increased epitaxy thickness $T_{\text{Si_epi}}$. The leakage path may be prevented by covering, during the selective epitaxy growth process, each region of the SOI wafer **100** in which a field oxide structure **118** is to be formed after the selective epitaxy growth process.

[0029] In FIG. **2B**, some of the one or more first regions **110** of the silicon device layer **102** covered by the hard mask **108** during the selectively growing of the epitaxial silicon **116** correspond to field oxide regions of lateral power semiconductor devices to be produced from the SOI wafer **100**. In FIGS. **2B-2C**, the part of the hard mask **108** and patterned resist **114** that cover an area of the SOI wafer **100** that corresponds to field oxide regions to be formed after the selective epitaxy growth process are labelled **108b** and **114b**, respectively. This is in addition to the hard mask and patterned resist structures shown in FIGS. **1B-1C** and which are labelled **108a** and **114a**, respectively, in FIGS. **2B-2C**. FIG. **2D** shows the SOI wafer **100** after the field oxide structure **118** are formed. The field oxide regions **118**, which are formed in the one or more first regions **110** of the silicon device layer **102** covered by the hard mask **108** during the selective epitaxy growth process, extend down to the buried oxide layer **106** to prevent the leakage path described above.

[0030] Some of the one or more first regions **110** of the silicon device layer **102** covered by the hard mask **108** during the selective epitaxy growth process may correspond to contact regions of power semiconductor devices to be formed in the silicon device layer **102** and/or to additional semiconductor devices having a lower breakdown voltage than the power semiconductor devices to be formed in the silicon device layer **102**, as explained above. Various embodiments of the one or more first regions **110** of the silicon device layer **102** covered by the hard mask **108** during the selective epitaxy growth process are described next.

[0031] FIG. **3** illustrates a plan view of a lateral power MOSFET or JFET produced from the SOI wafer **100**. The lateral power MOSFET or JFET is included in a die singulated from the SOI wafer **100**. The power MOSFET or JFET is a 'lateral' device in that the primary current flow path is along the front main surface of the silicon device layer **102** in the x direction, from the source **200** to the drain **202**. The source and drain **200**, **202** are of a first conductivity type. The body region of a lateral power MOSFET, which would be below the source region **200** and out of view in FIG. **3**, is of a second conductivity type opposite the first conductivity type. A body contact region **204** of the second conductivity type and having a higher doping concentration than the body region provides an ohmic connection to the overlying source metallization (not shown in FIG. **3**). The first conductivity is n-type and the second conductivity type is p-type for an n-channel device, whereas the first conductivity is p-type and the second conductivity type is n-type for a p-channel device.

[0032] A gate electrode **206** controls the conductive state of the device channel. In the case of a MOSFET, the gate electrode **206** is insulated from the silicon device layer **102**. In the case of a JFET, the gate electrode **206** contacts the silicon device layer **102**. The source region **200** is illustrated in a simplified manner in FIG. **3** but may be implemented as a plurality of islands. For a p-channel JFET, the source islands are p⁺ regions and the gate electrode **206** contacts n⁺ regions to pinch off the p-channel between two (2) neighboring source islands and thus control the on/off state of the JFET.

[0033] A drift zone **208** laterally extends between the gate electrode **206** and the drain region **202** of the lateral power MOSFET or JFET. The drift zone **208** may include a superjunction structure with oppositely doped first and second regions **210**, **212**, where the superjunction structure enables high blocking capability (e.g., >600V) without adversely affecting Rdson. The oppositely doped regions **210**, **212** of the superjunction structure may be 'stripe-shaped' in that the oppositely doped superjunction regions **210**, **212** have a longest linear dimension in the x direction in FIG. **3** and parallel to the front surface of the silicon device layer **102**. Four (4) superjunction stripes (2 of each conductivity type) are shown in FIG. **3**. However, the device may include more superjunction stripes.

[0034] As indicated by the dashed rectangle in FIG. **3**, the one or more second regions **112** of the silicon device layer **102** exposed by the hard mask **108** during the selective epitaxy growth process in FIGS. **1C** and **2C** correspond to the drift zone **208** of the lateral power MOSFET or JFET. Accordingly, the drift zone **208** of the lateral power MOSFET or JFET is thicker than the source, drain, body, and body contact regions **200**, **202**, **204** of the device. In the case of a superjunction device, this means that at least part of the oppositely doped first and second regions **210**, **212** of the superjunction structure formed in the drift zone **208** may be thicker than other regions of the lateral power MOSFET or JFET to reduce Rdson.

[0035] For example, the source, drain, body, and body contact regions **200**, **202**, **204** of the lateral power MOSFET or JFET may not benefit from an increased epitaxy thickness and therefore may correspond to the one or more first regions **110** of the silicon device layer **102** covered by the hard mask **108** during the selective epitaxy growth process in FIGS. **1C** and **2C**. As explained above, field oxide regions **118** used for isolation purposes may correspond to the one or more first regions **110** of the silicon device layer **102** covered by the hard mask **108** during the selective epitaxy growth process, e.g., as shown in FIG. **2C**, and therefore may extend down to the buried oxide layer **106** to prevent the leakage path described above.

[0036] In FIG. 3, the drift zone **208** of the device is devoid of contact regions. In this example, and during the selective epitaxy growth process shown in FIGS. 1C and 2C, the one or more second regions **112** of the silicon device layer **102** uncovered by the hard mask **108** expose the entirety of the drift zone area **208** between the source contact region **214** and the drain contact region **216** during the selectively growing of the epitaxial silicon **116** shown in FIGS. 1C and 2C. The source and drain contact regions **214**, **216** are illustrated as respective dashed rectangles in FIG. 3 since the contact regions **214**, **216** represent a vertical connection structure to the source and drain regions **200**, **202**, respectively. The source contact region **214** and the drain contact region **216** may include a single large contact (e.g., tungsten) or a row of smaller spaced apart contacts.

[0037] FIG. 4 illustrates a plan view of another embodiment of a lateral power MOSFET or JFET produced from the SOI wafer **100**. In FIG. 4, the drift zone **208** has contact regions **218** that are laterally spaced apart from one another along a length (x direction in FIG. 4) of the drift zone **208**. The drift zone contact regions **218** are illustrated as respective dashed rectangles in FIG. 4 since the drift zone contact regions **218** represent a vertical connection structure to the doped regions **212** of the superjunction structure having the second conductivity type. Each drift zone contact region **218** may include one or more contacts (e.g., tungsten) for each doped region **212** of the superjunction structure having the second conductivity type. In FIG. 4, the drift zone **208** is made thicker between the rows of the drift zone contact regions **218** but not below the rows of the contact regions **218** during the selective epitaxy growth process shown in FIGS. 1C and 2C. In this embodiment, each of the source contact regions **214**, the drain contact regions **216**, and the drift zone contact regions **218** have a thickness T_{Si_init} that remains unchanged by the selective epitaxy growth process shown in FIGS. 1C and 2C and correspond to the one or more first regions **110** of the silicon device layer **102** covered by the hard mask **108** during the selective epitaxy growth process.

[0038] FIG. 5 illustrates cross-sectional views of a low-voltage (e.g., <600V) lateral SOI MOS transistor **300** and a high-voltage (e.g., >600V) diode **302** that are monolithically integrated together in the same semiconductor die **304** using the SOI process described herein. In the case of an n-channel device, the low-voltage lateral SOI MOS transistor **300** has an n⁺ source **306**, a p body **308**, an n⁻ drift zone **310**, and an n⁺ drain **312** formed in the silicon device layer **102**. A gate electrode **314** insulated from the p body **308** by a gate dielectric **315** controls the channel in the p body region **308**. A p-type shield well **316** is formed in the bulk silicon substrate **104** and is electrically contacted by a contact **318** that extends through the buried oxide layer **106**. A source contact **320** similarly electrically contacts the n⁺ source region **306** and a drain contact **322** similarly electrically contacts the n⁺ drain region **312**.

[0039] The high-voltage diode **302** includes a p anode region **324** formed in the bulk silicon substrate **104** and that is electrically contacted by a contact **326** that extends through the buried oxide layer **106**. P-type shielding rings **328** and an n-type edge termination doping region **330** also may be formed in the bulk silicon substrate **104**. The n⁺cathode region **332** is formed at the backside of the bulk silicon substrate **104**.

[0040] The monolithically integrated semiconductor die **304** includes both the low-voltage lateral SOI MOS transistor **300** and the high-voltage diode **302** formed in the same silicon device layer **102**. The n⁻ drift zone **310** of the low-voltage lateral SOI MOS transistor **300** may be lengthened (x direction in FIG. 5) and thickened (z direction in FIG. 5) to accommodate a higher blocking voltage capability. The n⁻ drift zone **310** may be made thicker (T_{Si_epi}) by unmasking the corresponding region of the silicon device layer **102** during the selective epitaxy growth process shown in FIGS. 1C and 2C. The thickness (T_{Si_init}) of other regions of the silicon device layer **102** may remain unchanged during the selective epitaxy growth process shown in FIGS. 1C and 2C, by masking these regions of the silicon device layer **102** during the selective epitaxy growth process. For example, the regions of the silicon device layer **102** that correspond to the source, drain, gate, and contact regions **306**, **312**, **314**, **318**, **320**, **322**, **334** of the monolithically integrated device may be masked during the selective epitaxy growth process shown in FIGS. 1C and 2C.

[0041] FIG. 6 illustrates a plan view of an embodiment of a power transistor device **400** included in a semiconductor die produced using the SOI process described herein. Two (2) gate fingers **402** are shown in FIG. 6, but the power transistor device **400** may include more than two gate fingers **402**. Each gate finger **402** controls a conduction channel between a group of source islands **404** of the first conductivity type and a drift zone **406** of the first conductivity type, through a corresponding body region **408** of the second conductivity type. A body contact region **410** of the second conductivity type and having a higher doping concentration than the body region provides an ohmic connection to the overlying source metallization (not shown in FIG. 3). The drift zones **406** extend to a drain region **412** of the first conductivity type. Source/body and drain contact regions **414**, **416** are illustrated as respective dashed rectangles in FIG. 6 since the contact regions **414**, **416** represent a vertical connection structure to the source/body and drain regions **404/408**, **412**, respectively, of the power transistor device **400**.

[0042] For the power transistor device **400** shown in FIG. 6 and other transistor devices with a similar layout, the entire active area may be uncovered by the hard mask **108** during the selective epitaxy growth process shown in FIGS. 1C and 2C, since a thin (T_{Si_init}) SOI area is not required for any region of the device. The large rectangle surrounding the active area in FIG. 6 corresponds to a second region **112** of the silicon device layer **102** that is uncovered by the hard mask **108** during the selectively growing of the epitaxial silicon **116** shown in FIGS. 1C and 2C. Accordingly, the thickness of the entire active area of the power transistor device **400** is increased from T_{Si_init} to T_{Si_epi} by the selective epitaxy growth process shown in FIGS. 1C and 2C. The one or more first regions **110** of the silicon device layer **102** protected by the hard mask **108** during the selective epitaxy growth process shown in FIGS. 1C and 2C may correspond to an additional device (not shown) monolithically integrated with the power transistor device **400** on the same semiconductor die and having a lower breakdown voltage than the power transistor device **400**.

[0043] FIG. 7 illustrates a plan schematic view of a semiconductor die **500** that includes one or more low-voltage (e.g., breakdown voltage <600V) areas **502** and one or more high-voltage (e.g., breakdown voltage >600V) areas **504**. The large rectangle surrounding each high-voltage area **504** in FIG. 6 corresponds to a second region **112** of the silicon device layer **102** that is uncovered by the hard mask **108** during the selectively growing of the epitaxial silicon **116** shown in FIGS. 1C and 2C. Accordingly, the thickness of the silicon device layer **102** in the one or more high-voltage areas **504** increases from T_{Si_init} to T_{Si_epi} during the selective epitaxy growth process shown in FIGS. 1C and 2C. The thickness of the silicon device layer **102** may be increased to T_{Si_epi} throughout the entire active area of the one or more high-voltage areas **504**, e.g., as shown in FIG. 6, or in a select region(s) such as the drift zone, e.g., as shown in FIGS. 3 and 4 where other regions like contact regions retain the initial S thickness T_{Si_init}.

[0044] Each low-voltage area **502** is protected by the hard mask **108** during the selective epitaxy growth process shown in FIGS. 1C and 2C and therefore retains the initial Si device layer thickness T_{Si_init} throughout. The high-voltage area(s) **504** may include power semiconductor devices such as a high-voltage diode, a lateral power MOSFET, a lateral JFET with a high-voltage NMOS device, etc. The low-voltage area(s) **502** may include, e.g., logic devices, low-voltage NMOS devices, driver circuitry, or other types of semiconductor devices having a lower breakdown voltage than the power semiconductor devices included in the one or more high-voltage areas **504**.

[0045] Although the present disclosure is not so limited, the following numbered examples demonstrate one or more aspects of the disclosure.

[0046] Example 1. A method of producing power semiconductor devices from a silicon-on-insulator (SOI) wafer that includes a silicon device layer, a bulk silicon wafer, and a buried oxide layer separating the silicon device layer from the bulk silicon wafer, the method comprising: forming a hard mask on the silicon device layer, wherein the hard mask covers one or more first regions of the silicon device layer and exposes one or more second regions of the silicon device layer; and before forming any field oxide structures and before implanting any device regions,

selectively growing epitaxial silicon on the one or more second regions of the silicon device layer exposed by the hard mask such that the thickness of the one or more second regions is increased relative to the one or more first regions.

[0047] Example 2. The method of example 1, wherein the power semiconductor devices are lateral power MOSFETs (metal-oxide-semiconductor field-effect transistors), and wherein the one or more second regions of the silicon device layer exposed by the hard mask during the selectively growing of the epitaxial silicon correspond to drift zones of the lateral power MOSFETs.

[0048] Example 3. The method of example 2, wherein the one or more first regions of the silicon device layer covered by the hard mask during the selectively growing of the epitaxial silicon correspond to contact regions of the lateral power MOSFETs.

[0049] Example 4. The method of example 3, wherein a plurality of rows of the contact regions are laterally spaced apart from one another along a length of the drift zones, and wherein during the selectively growing of the epitaxial silicon, the drift zones are made thicker between the rows of the contact regions but not below the rows of the contact regions.

[0050] Example 5. The method of example 3 or 4, wherein a first row of the contact regions are source contact regions having a thickness that remains unchanged by the selectively growing of the epitaxial silicon, and wherein a second row of the contact regions are drain contact regions having a thickness that remains unchanged by the selectively growing of the epitaxial silicon.

[0051] Example 6. The method of example 1, wherein the power semiconductor devices are lateral JFETs (junction field-effect transistors), and wherein the one or more second regions of the silicon device layer exposed by the hard mask during the selectively growing of the epitaxial silicon correspond to drift zones of the lateral JFETs.

[0052] Example 7. The method of example 6, wherein the one or more first regions of the silicon device layer covered by the hard mask during the selectively growing of the epitaxial silicon correspond to source contact regions and drain contact regions of the lateral JFETs, and wherein both the source contact regions and the drain contact regions have a thickness that remains unchanged by the selectively growing of the epitaxial silicon.

[0053] Example 8. The method of example 7, wherein the drift zones are devoid of contact regions, and wherein the one or more second regions of the silicon device layer expose an entirety of the drift zone area between the source contact regions and the drain contact regions during the selectively growing of the epitaxial silicon.

[0054] Example 9. The method of any of examples 1 through 8, wherein the one or more second regions of the silicon device layer exposed by the hard mask during the selectively growing of the epitaxial silicon correspond to drift zones of the power semiconductor devices, and wherein the one or more first regions of the silicon device layer covered by the hard mask during the selectively growing of the epitaxial silicon correspond to additional semiconductor devices having a lower breakdown voltage than the power semiconductor devices.

[0055] Example 10. The method of any of examples 1 through 9, wherein the one or more first regions of the silicon device layer covered by the hard mask during the selectively growing of the epitaxial silicon correspond to field oxide regions of the power semiconductor devices.

[0056] Example 11. The method of any of examples 1 through 10, wherein the one or more first regions of the silicon device layer covered by the hard mask during the selectively growing of the epitaxial silicon correspond to contact regions of the power semiconductor devices.

[0057] Example 12. The method of any of examples 1 through 11, wherein the power semiconductor devices are PMOS devices having a breakdown voltage greater than 600V, and wherein the one or more first regions of the silicon device layer covered by the hard mask during the selectively growing of the epitaxial silicon correspond to additional semiconductor devices having a breakdown voltage less than 600V.

[0058] Example 13. The method of any of examples 1 through 12, wherein before the selectively growing of the epitaxial silicon, the silicon device layer has a single thickness in a range of 100 to

200 nm.

[0059] Example 14. The method of any of examples 1 through 13, wherein the hard mask comprises tetraethoxysilane formed by low-pressure chemical vapor deposition.

[0060] Example 15. A semiconductor die, comprising: a silicon-on-insulator (SOI) substrate that includes a silicon device layer, a bulk silicon substrate, and a buried oxide layer separating the silicon device layer from the bulk silicon substrate; a lateral power MOSFET (metal-oxide-semiconductor field-effect transistor) formed in a first device region of the silicon device layer; and an additional semiconductor device formed in a second device region of the silicon device layer and having a lower breakdown voltage than the lateral power MOSFET, wherein the silicon device layer has a first thickness in a first part of the first device region and a second thickness in a second part of the first device region, the second thickness being greater than the first thickness, wherein the silicon device layer has the first thickness throughout the second device region.

[0061] Example 16. The semiconductor die of example 15, wherein the second part of the first device region includes a drift zone of the lateral power MOSFET, and wherein the first part of the first device region includes contact regions of the lateral power MOSFET.

[0062] Example 17. The semiconductor die of example 16, wherein the contact regions are arranged in rows that are laterally spaced apart from one another along a length of the drift zone such that the drift zone is thicker between the rows of the contact regions and thinner below the rows of the contact regions.

[0063] Example 18. The semiconductor die of example 16 or 17, wherein a first row of the contact regions are source contact regions of the lateral power MOSFET and a second row of the contact regions are drain contact regions of the lateral power MOSFET, and wherein the drift zone is devoid of the contact regions such that the drift zone has the second thickness along an entirety of the drift zone area between the source contact regions and the drain contact regions.

[0064] Example 19. The semiconductor die of example 15, wherein the second part of the first device region includes a drift zone of the lateral power MOSFET, and wherein the first part of the first device region includes field oxide regions.

[0065] Example 20. A semiconductor die, comprising: a silicon-on-insulator (SOI) substrate that includes a silicon device layer, a bulk silicon substrate, and a buried oxide layer separating the silicon device layer from the bulk silicon substrate; and a lateral JFET (junction field-effect transistor) formed in the silicon device layer, wherein a first part of the silicon device layer has a first thickness and a second part of the silicon device layer has a second thickness greater than the first thickness, wherein, in the silicon device layer, the lateral JFET comprises a source region, a drain region, a drift zone adjoining the drain region, and a body region separating the source region from the drift zone, wherein the drift zone is formed in the second part of the silicon device layer.

[0066] Example 21. The semiconductor die of example 20, wherein contact regions of the lateral JFET are formed in the first part of the silicon device layer.

[0067] Example 22. The semiconductor die of example 21, wherein a first row of the contact regions are source contact regions of the lateral JFET and a second row of the contact regions are drain contact regions of the lateral JFET, and wherein the drift zone is devoid of the contact regions such that the drift zone has the second thickness along an entirety of the drift zone area between the source contact regions and the drain contact regions.

[0068] Terms such as “first”, “second”, and the like, are used to describe various elements, regions, sections, etc. and are also not intended to be limiting. Like terms refer to like elements throughout the description.

[0069] As used herein, the terms “having”, “containing”, “including”, “comprising” and the like are open ended terms that indicate the presence of stated elements or features, but do not preclude additional elements or features. The articles “a”, “an” and “the” are intended to include the plural as well as the singular, unless the context clearly indicates otherwise.

[0070] It is to be understood that the features of the various embodiments described herein may be

combined with each other, unless specifically noted otherwise.

[0071] Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations may be substituted for the specific embodiments shown and described without departing from the scope of the present invention. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. Therefore, it is intended that this invention be limited only by the claims and the equivalents thereof.

Claims

1. A semiconductor die, comprising: a silicon-on-insulator (SOI) substrate that includes a silicon device layer, a bulk silicon substrate, and a buried oxide layer separating the silicon device layer from the bulk silicon substrate; a lateral power MOSFET (metal-oxide-semiconductor field-effect transistor) formed in a first device region of the silicon device layer; and an additional semiconductor device formed in a second device region of the silicon device layer and having a lower breakdown voltage than the lateral power MOSFET, wherein the silicon device layer has a first thickness in a first part of the first device region and a second thickness in a second part of the first device region, the second thickness being greater than the first thickness, wherein the silicon device layer has the first thickness throughout the second device region.
2. The semiconductor die of claim 1, wherein the second part of the first device region includes a drift zone of the lateral power MOSFET, and wherein the first part of the first device region includes a plurality of contact regions of the lateral power MOSFET.
3. The semiconductor die of claim 2, wherein the contact regions are arranged in rows that are laterally spaced apart from one another along a length of the drift zone such that the drift zone is thicker between the rows of the contact regions and thinner below the rows of the contact regions.
4. The semiconductor die of claim 2, wherein a first row of the contact regions are source contact regions of the lateral power MOSFET and a second row of the contact regions are drain contact regions of the lateral power MOSFET, and wherein the drift zone is devoid of the contact regions such that the drift zone has the second thickness along an entirety of the drift zone area between the source contact regions and the drain contact regions.
5. The semiconductor die of claim 2, wherein the drift zone includes a superjunction structure comprising oppositely doped first and second regions formed in the second part of the first device region.
6. The semiconductor die of claim 5, wherein the oppositely doped first and second regions are stripe-shaped and have a longest linear dimension in a direction parallel to a front surface of the silicon device layer.
7. The semiconductor die of claim 5, wherein the oppositely doped first and second regions are thicker than source, drain, and body regions formed in the first part of the first device region.
8. The semiconductor die of claim 2, wherein the drift zone is thicker than source, drain and body regions formed in the first part of the first device region.
9. The semiconductor die of claim 1, wherein the second part of the first device region includes a drift zone of the lateral power MOSFET, and wherein the first part of the first device region includes field oxide regions.
10. The semiconductor die of claim 1, further comprising: a gate electrode configured to control a conductive state of a channel of the lateral power MOSFET, wherein the gate electrode is insulated from the silicon device layer.
11. The semiconductor die of claim 1, further comprising: a p-type shield well formed in the bulk silicon substrate and electrically contacted by a contact that extends through the buried oxide layer.
12. The semiconductor die of claim 1, further comprising: a plurality of p-type shielding rings and an n-type edge termination doping region formed in the bulk silicon substrate.

- 13.** A semiconductor die, comprising: a silicon-on-insulator (SOI) substrate that includes a silicon device layer, a bulk silicon substrate, and a buried oxide layer separating the silicon device layer from the bulk silicon substrate; and a lateral JFET (junction field-effect transistor) formed in the silicon device layer, wherein a first part of the silicon device layer has a first thickness and a second part of the silicon device layer has a second thickness greater than the first thickness, wherein, in the silicon device layer, the lateral JFET comprises a source region, a drain region, a drift zone adjoining the drain region, and a body region separating the source region from the drift zone, wherein the drift zone is formed in the second part of the silicon device layer.
- 14.** The semiconductor die of claim 13, wherein a plurality of contact regions of the lateral JFET are formed in the first part of the silicon device layer.
- 15.** The semiconductor die of claim 14, wherein a first row of the contact regions are source contact regions of the lateral JFET and a second row of the contact regions are drain contact regions of the lateral JFET, and wherein the drift zone is devoid of the contact regions such that the drift zone has the second thickness along an entirety of the drift zone area between the source contact regions and the drain contact regions.
- 16.** The semiconductor die of claim 13, wherein the drift zone includes a superjunction structure comprising oppositely doped first and second regions formed in the second part of the silicon device layer.
- 17.** The semiconductor die of claim 16, wherein the oppositely doped first and second regions are stripe-shaped and have a longest linear dimension in a direction parallel to a front surface of the silicon device layer.
- 18.** The semiconductor die of claim 16, wherein the oppositely doped first and second regions are thicker than the source region, the drain region and the body region.
- 19.** The semiconductor die of claim 13, wherein the drift zone is thicker than the source region, the drain region and the body region.
- 20.** The semiconductor die of claim 13, further comprising: a gate electrode configured to control a conductive state of a channel of the lateral JFET, wherein the gate electrode contacts the silicon device layer.
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